

More than an empty case: a non invasive technique for monitoring the Australian critically endangered golden sun moth, *Synemon plana* (Lepidoptera: Castniidae)

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Abstract Monitoring programs for butterflies and moths are focused on the adult stage, rarely considering other life stages. Transect-based counts of adults and searches for empty pupal cases have been suggested as standard monitoring protocols for the critically endangered golden sun moth *Synemon plana* in Australia. To date, surveys and monitoring have focused only on counts of adults. However, undertaking such counts is constrained by the short adult life of the species (1–2 days), and the fact that prevailing weather conditions can seriously influence detectability. We tested whether empty pupal cases of *S. plana* can be used to supplement the monitoring of adults and whether this technique can be undertaken by citizen scientists. Volunteers from Canberra (Australia) collected 650 pupal cases from 11 grassland areas. The cases were found in native grasslands and in grassland comprised entirely of the exotic Chilean needle grass (*Nassella neesiana*). Pupal cases of *S. plana* were found to be durable, with most persisting in the field for greater than 3 weeks after first sighting, and exhibited a male biased sex ratio. This study demonstrates that detection of empty pupal cases provides a potential additional tool to monitor *S. plana* that is not dependent on the restrictions of prevailing weather

conditions and time of day, and can be undertaken by citizen scientists.

Keywords Golden sun moth · *Synemon plana* · Monitoring · Pupal cases · Sex ratio · Citizen scientists

Introduction

Monitoring is a powerful tool in conservation biology because it provides a measure of population characteristics (e.g. relative abundance, demographic characteristics) that can be collected over time and used to determine relationships with environmental factors such as climate, land use, fragmentation, harvesting and management (Lindenmayer and Likens 2010; Spellerberg 2005). Butterflies and moths are highly suited as environmental indicators and have attracted considerable public interest as groups suitable for long term monitoring (Van Swaay et al. 2008). For example, information derived from European butterfly monitoring programs over the past 30 years have provided evidence of a significant decline in many species, and evidence of an extension of the range of some species, principally towards northerly latitudes (Asher et al. 2011; Van Swaay et al. 2008). The application of monitoring data has led to improved conservation practices in many countries and this has halted the decline of some species (Asher et al. 2011; Brereton et al. 2011). Although changes in methods can confound the interpretation of results derived from monitoring programs, such programs are often not static and may be modified to improve statistical power or to improve the efficiency of the techniques used (Lindenmayer and Likens 2010). For example, low population numbers can lead to insufficient data, particularly for rare and endangered species. Species specific characteristics

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such as high mobility or cryptic behavior can constrain the detectability of species and consequently lead to bias towards the more obvious and common species (VanStrien et al. 1997). Prevailing weather conditions are often not suitable for monitoring weather sensitive groups, such as butterflies and moths, and this presents a challenge to volunteers and professionals.

Butterfly and moth monitoring programs conducted with the assistance of citizen scientists generally focus on the adult stage. This is because monitoring adult life forms is typically more appealing to volunteers than recording non adult life forms (Nowicki et al. 2008; Van Swaay et al. 2008). Monitoring non adult stages is potentially of considerable importance for species with very short flight phenologies or for species that are difficult to detect. Monitoring of larvae is a key component of the monitoring program for the iconic monarch butterfly (*Danaus plexipus*) in North America (Garland and Davis 2002; Gibbs et al. 2006) and for lycaenid and fritillary butterflies in Europe (Maes et al. 2004; Konvicka et al. 2005). In Australia, where there are few established monitoring programs for Lepidopterans, several programs include the monitoring of non adult life forms (New 2010). For example, long-term monitoring of the Eltham Copper Butterfly (*Paralucia pyrodiscus lucida*) has included nocturnal counts of larvae (Braby et al. 1999; see New 2010 for review). Also, monitoring activities for the Australian threatened Richmond Birdwing (*Ornithoptera richmondia*) include recording larvae and pupae (Sands 2008). Searches for pupal cases have been recommended as one component of surveys for the critically endangered golden sun moth (*S. plana*) (Australian Government 2009). If non adult forms of *S. plana* can be cost-effectively measured by citizen scientists, then the inclusion of these alternative life forms in monitoring programs will provide valuable information. To examine this possibility we tested whether the collection of pupal cases of the golden sun moth (*S. plana*) can be used as a supplement to the adult surveys. Furthermore, we assessed if this approach can be conducted with the assistance of citizen scientists. To do this we trained volunteers to systematically collect the pupal cases at selected sites in the Canberra region (Richter et al. 2009). We also examined the survival of the pupal cases when left in situ in the field and used those that were collected to determine the sex-ratio of the species at emergence.

The study species

The golden sun moth is listed as critically endangered nationally under the Commonwealth Environment Protection and Biodiversity Conservation Act (1999) (from 2002) and listed as threatened in Victoria (Flora and Fauna

Guarantee Act 1998), endangered in the Australian Capital Territory (Nature Conservation Act 1980, listed in 2006) and endangered in New South Wales (Threatened Species Conservation Act 1995; listed in 1996). The original natural habitat of the species in south-eastern Australia is lowland natural temperate grassland, an ecological community that occurs in a naturally treeless landscape often located in frost hollows and on open plains. Sites supporting the moth are characterised by the presence of native grasses including species of *Austrodanthonia* (Edwards 1994). Since European settlement more than 95 % of these grasslands have been destroyed or highly modified as a result of agricultural, rural and urban development (Kirkpatrick et al. 1995). *Synemon plana* is likely to have been widely distributed across these native grasslands at the time of European settlement (Clarke and Whyte 2003), but now has been largely extirpated throughout its former range (Braby and Dunford 2006; Clarke and O'Dwyer 2000). When our study commenced in 2007 the species was known from 125 sites (records post 1990) across a potential range of more than two million hectares. Recent surveys in Victoria have almost doubled the number of known sites (Brown et al. 2012). Most of the grassland sites where the species has been recorded are smaller than five hectares and face ongoing pressures from rapid urban and rural expansion and degradation (Braby and Dunford 2006; Gibson and New 2007; Gilmore et al. 2008).

The adults are diurnal and have a flight season between mid October and early January (spring to summer), depending on local climate and topography. The species is only active during the hottest part of the day (1000–1500 hours) (possibly up to 1700 hours for females) on sunny and cloudless days with a low to moderate wind speed. Temperatures can reach 37 degrees Celsius during these conditions. In the field, adult golden sun moths typically live for 1–2 days (Richter 2010) and lack functional mouthparts for feeding or ingesting any liquid. Females are rarely observed to fly, and when they do the distances traveled in a single flight are typically in the order of a few metres (up to 20 m), whereas males have been observed to fly for several hundred metres. Males fly in a zigzag pattern above the grass sward to search for receptive females; the females remain on the ground and display their bright orange hind wings to attract males. Immediately after mating females crawl quickly amongst the grass tussocks and lay between 80 and 100 eggs at the base of native grass tussocks (Edwards 1994). Based on our observations of oviposition, the eggs are not clumped but appear to be spread over many tussocks. Larvae are thought to emerge about 20 days after laying and feed on the bases of the culms and roots of native grasses (Edwards 1993). Based on the size classes of larvae encountered in the soil it appears that they remain underground for at least 2 years

(Edwards 1994). Pupation occurs in spring. The larvae prepare a burrow (feeding tube) that leads to the soil surface and move to the ground surface where they pupate and emerge. Following adult emergence the pupal shells are left partially protruding above the surface or fully exposed. Pupal cases can be readily found by searching through and around the shorter tussocks and inter-tussock spaces (Richter et al. 2009).

The presence of *S. plana* populations and their pupal cases in natural temperate grassland dominated by *Austrodanthonia* has led to the assumption that the species is a dietary specialist, feeding only on selected species in this genus (most notably *Austrodanthonia carphoides*) (Edwards 1993). Recently, several *S. plana* populations have been recorded at sites dominated by the exotic grass Chilean needle grass (*Nassella neesiana*) (Braby and Dunford 2006; Gilmore et al. 2008) but the suitability of this grass as a component of the diet of the moths needs further investigation. Chilean needle grass is native to South America, was introduced to Melbourne, Australia in 1934 and from there has spread into many locations throughout Australia. It is widespread in parts of the cities of Melbourne and Canberra and was declared a Weed of National Significance in Australia (Anonymous 1999). The assessment of the current distribution of *S. plana* in native and Chilean needle grass dominated grasslands has been constrained by: (1) the bias in counts of adult *S. plana* towards males because of their more conspicuous flight behavior; (2) the difficulty of collecting data due to the restricted time that adult surveys can be conducted because of frequent unsuitable weather; (3) the difficulty of collecting data from several populations at the same time because of the short daily activity period; and (4) the fact that an absence of adult moths can only be inferred if observations are made under ideal weather conditions. Furthermore, the lack of tested monitoring methods exacerbates comparative analysis.

Methods

The study was undertaken between September 2007 and April 2008 in and near Canberra, Australia. The research was conducted with the help of volunteer citizen scientists participating in the ‘Sun Moth Count’ project coordinated by the Institute for Applied Ecology at the University of Canberra (Richter et al. 2009). More than 50 members of the community took part in workshops, training sessions and the on-site monitoring program. Four workshops of more than 6 h each and three field training sessions were held to introduce the participants to monitoring techniques for *S. plana*. On-site guidance was provided to volunteers throughout the project. Each volunteer was assigned to a particular grassland site. The 11 survey locations were grouped into native grassland sites (grassland dominated by

native grasses e.g. *Austrodanthonia* spp, *Austrostipa* spp and forbs) and exotic grasslands (in this case grassland entirely dominated by the invasive Chilean Needle grass [*N. neesiana*]). For each location we prepared a digital map that showed the specific positions where 12 randomly selected 1 m² survey plots were to be placed. Each plot was visited four times (no closer together than 1 week) before the survey was considered to be finished. During each visit we asked the participants to search for, and collect, all empty *S. plana* pupal cases found in the survey plots. To prevent damage to the cases they were often removed with the soil and silk webbing that was attached to them. These were later cleaned and sexed by comparison with a reference collection of male and female pupal cases. The total length of each case was also measured with digital calipers.

At one representative site in natural temperate grassland in the suburb of Belconnen (former Belconnen Naval Station), 31 *S. plana* pupal cases were left in situ and their locations individually marked with flags. These cases were examined from 7 November 2008 to 12 January 2009 to estimate their durability under natural conditions (degradation by wind or rain action or trampling by kangaroos). Daily weather conditions (rainfall and temperature) during the study were derived from data provided by the Bureau of Meteorology for the nearest weather station at Mitchell (3.5 km away). For the analysis of the durability of the pupal cases we used the program MARK to calculate the survival probability (ϕ) for an open population (assemblage of pupal cases) where birth (emergences of new cases) and death (degradation of cases) can occur. One-way and two-way factorial analysis of variance (ANOVA) using generalized linear models (GLM SPSS) were used to test for significant differences in sex ratios among types of grassland and in sizes of pupal cases.

Results

A total of 650 pupal cases were collected of which 479 were obtained from eight locations in native temperate grassland (NTG) and 171 from three locations in grasslands that were dominated by the exotic Chilean needle grass (CNG). Less than 2 % of all pupal cases collected by the citizen scientists had been misidentified. The mean female pupal case length was 23.15 mm ($n = 145$, 0.18 SE) whereas the mean male pupal case length was 21.66 mm ($n = 238$, 0.24 SE) and these differences were significant ($F = 11.451$, $df = 1,377$, $p = 0.001$). Cases collected from exotic grassland (Chilean needle grass) were significantly larger than cases collected in natural temperate grassland ($F = 30.299$, $df = 2,377$, $p = 0.000$). The ratio of males to females in *S. plana* populations in natural temperate grassland was found to vary between 0.6 and 3.5

with a mean 1.9 sex ratio (Table 1). Also, nearly twice as many *S. plana* males than females were found at sites dominated by Chilean needle grass (Table 1). The sex ratio bias towards males did not vary between the 2 years studied (Fig. 1). The survival analysis performed in program MARK (version 5.1) on 31 empty pupal cases showed that *S. plana* pupal cases had a high “survival” rate ($\phi = 0.9\text{--}1.0$) between one and 24 days after marking (Fig. 2). A large proportion of the empty cases (95 %) could still be found after 24 days, and this was after an extreme rainfall event on 13 November 2008 when 60 mm of rain were recorded (Table 2). Their detectability decreased after 24 days, and after more than 40 days < 50 % of *S. plana* cases were still present and identifiable in the field. At the last visit to check for the presence of empty cases 50 days after initial marking, all cases had degraded.

Discussion

The survey and monitoring guidelines for *S. plana* recommend fixed point (or “spot count”) and transect surveys as standard tools for the detection of adult *S. plana* (DEWHA 2009). The guidelines also recommend the use of pupal case surveys, however the usefulness of this as an alternative monitoring tool has not been assessed and we are not aware of anyone employing this approach. Our findings indicate that searches for empty pupal cases can provide a potential additional tool for monitoring *S. plana* populations that is not dependent on prevailing weather conditions and time of day. We demonstrated that monitoring empty pupal cases can be undertaken readily by citizen scientists (trained volunteers) and that examination

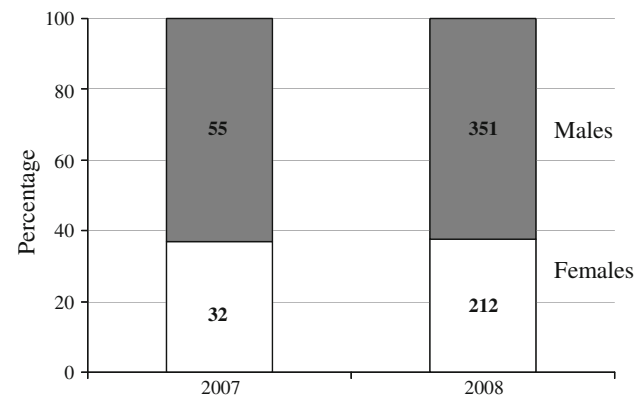


Fig. 1 Sex ratio in *S. plana* populations based on pupal case identification from samples ($n = 650$) obtained in 2007 and 2008 in native temperate grassland and exotic grassland dominated by Chilean needle grass (*N. neesiana*). Males are indicated by shaded bars, females by open bars. The number of specimens examined in each year is listed within the bars

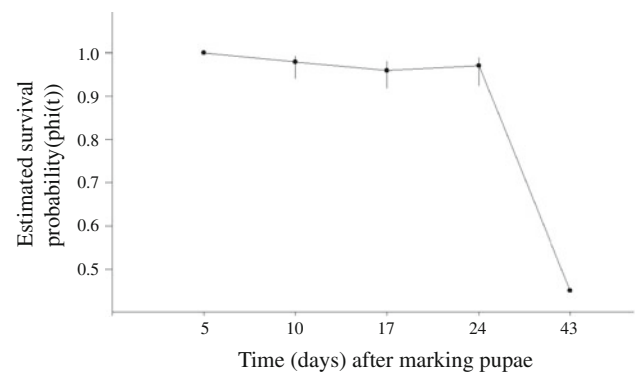


Fig. 2 Pupal case durability under natural conditions based on the survival probability analysis [$\phi(t)$] between 07.11.2008 and 12.01.2009 at one of the largest natural temperate grasslands in the ACT (former Belconnen Naval Transmitting Station). Bars indicate standard error

Table 1 Sex ratio variations in *S. plana* pupal cases (exuviae) collected from Chilean needle grass sites (A–C) and native grassland sites (1–8) in the Australian Capital Territory (ACT)

Site	Vegetation classification	Male: female	Sample size
A	CNG	2.5	14
B	CNG	2.1	114
C	CNG	1.9	43
1	NTG	3.5	9
2	NTG	2.5	14
3	NTG	2.4	111
4	NTG	1.8	91
5	NTG	1.3	7
6	NTG	1.3	214
7	NTG	0.6	8
8	NTG	0.6	25

of the cases provides information about the occurrence of the species and additional biological information.

Most international monitoring programs for butterflies and moths focus on recording the presence and abundance of adults. Recording the response of other life stages to changing environments has received less attention (Nowicki et al. 2008). This bias is not surprising given that most species are easily recognised as adults, and in many countries the biology and ecology of most species are reasonably well documented. This has allowed for the development of reliable approaches to monitoring. Despite the appeal of focusing on the adult life stage, most species of insects spend the greater part of their life either as an egg, larva or pupa. It is obvious that climate change, habitat destruction and inappropriate management will also impact on individuals at these early life stages (Smart et al. 2000; WallisDeVries et al. 2011). Therefore, documentation of

Table 2 Total rainfall (mm) and average maximal temperature (grad Celsius) occurring during the test of durability of empty pupal cases

Rainfall (past 24 h)	mm		
13 Nov 2008	62.2		
18 Nov 2008	6.2		
23 Nov 2008	1.4		
24 Nov 2008	4.4		
28 Nov 2008	24.6		
Average max temperature (past 24 h)	Celsius	Min.	Max.
07 Nov–30 Nov 2008	23.9	13	28.9
01 Dec–31 Dec 2008	25.4	19.8	30.7
01 Jan–12 Jan 2009	28.5	21.9	36.5

the responses of particular environmental perturbations to non-adult life forms is also relevant for the assessment of their conservation status. Unfortunately, the comparative ease of working with adult butterflies and moths has factored against the development of techniques for non-adult life stages. Nevertheless, for species with very short adult phenology or for species where the detection of adults is difficult (e.g. cryptic behaviour, flying high in trees), the identification of egg, larval or pupal cases can be straightforward and even in some cases advantageous over the identification of the adults (Doeringhaus et al. 2005; Hermann 1999; Lewis and Hurford 1997).

We have demonstrated that counts of pupal cases of *S. plana* can be used as a valuable tool for monitoring local populations. The practical factors that enhance the value of this approach include the following. First, since individual adult golden sun moths only survive for a day or two, observations of the adults can vary considerably from day to day. Such variation should not be present when pupal cases are counted (unless the site is disturbed in some way during the census period). If the counts are directed towards the end of the flight season then we expect that they will provide a reasonable estimate of accumulated population density for the entire season. By counting on multiple plots, it should be possible to calculate a mean value for emergence density of moths in that season, allowing for comparison with other sites. Second, the monitoring of pupal cases is independent of the species flight season. We demonstrated that pupal cases of *S. plana* are still present and identifiable under natural conditions 3 weeks after first detection (Fig. 2). In fact, the pupal cases ‘survived’ several days of high temperatures, rainfall and trampling from herbivores (kangaroos) (Table 2). By contrast, adults can be difficult to detect because of their short period of daily activity, the narrow range of suitable weather conditions that enable emergence and flight and their short life span. Thus, the monitoring of the presence and density of adult *S. plana* can be

supplemented by the documentation of the presence and density of empty pupal cases. Our study demonstrates that monitoring that is based on counting pupal cases can start after the first emergence of adults and can continue until beyond the end of the flight period. The durability of pupal cases should be considered when establishing a suitable interval between counts. An overestimation of the density of the moth through multiple counts of the same cases can be avoided by marking them with paint or removing the cases. At this stage, however, the relationship between the total population of adults that have emerged by a given date and the accumulated total number of pupal cases that can be found is not known and requires further research. We note that most grassland sites surveyed during our study were characterised by short vegetation as a result of severe drought conditions. Therefore, a possible limitation of the approach is that it might be difficult to find and count the pupal cases in situations where the grass is taller—such as might occur during wetter growing seasons. Finally we note and agree with the caution given by New (2012) that the immature stages of *S. plana* have not been formally described and that specialists are required for correct identification. This consideration needs to be factored into any surveys that are based on counts of pupal cases.

Monitoring programs document trends in the distribution of species but also provide valuable information about the biology and ecology of the species. Many butterfly species live in close association with plants (host plant relationship) and interact with other animals (e.g. ants) (New 1993). Thus, the observation of eggs and larvae on host plants, and any records of interactions with other taxa (e.g. larvae being fed by ants); indicate a well functioning relationship between butterfly and moth species and their environment. This information can be used as a supplement for the assessment of the species conservation status at a local scale.

Our study has provided the first record of the sex ratio of emerging adult *S. plana* (based on the sex of pupal cases). The fact that the ratio of reproductive males to females differed from parity in both native and exotic grassland habitats was unexpected. Generally, the operational sex ratio (OSR) is a central life history parameter that affects the population’s growth rate and determines the strength of sexual selection through competition for access to mates of the minority sex (Emlen and Oring 1977). When reared in captivity, most species of lepidopterans have an unbiased primary sex ratio (offspring sex ratio at the time of conception) regardless of their sex determining system (Ehrlich et al. 1984). However, it has been observed that some species of butterflies exhibit variable sex ratios under natural conditions (Dyson and Hurst 2004). Deviation from parity is likely to be corrected through frequency dependent selection, which acts to equalize parental investment

in the two sexes (Fisher 1930). Such adjustment arises through the fitness advantage incurred by the individuals that are genetically predisposed to bias their brood towards the minority sex, which in turn are more likely to encounter a mate than members of the majority sex and therefore provide a higher reproductive return and fitness gain to their parents. We detected a distinctive male-biased sex ratio in *S. plana* populations that was consistent across 2 years and did not differ significantly between native grassland and exotic *N. neesiana* grassland (Fig. 2). The factors that are likely to maintain a male bias in *S. plana* populations are unknown at this stage. To further understand the conditions that facilitate such dynamics, both physiological and ecological parameters of the species life history must be considered. For instance, sex determination might be influenced by environmental factors such as egg incubation temperature, which in turn could account for some or all of the observed bias. Species that produce a male bias are common, but so are those that produce equal sex ratios or only female offspring (Adamski 2004; Jiggins et al. 1998). The underlying causes of such variations in the primary sex ratios are unknown but suggested causes for this have included maternal inheritance (Jiggins et al. 1998), biased predation as a consequence of sexual dimorphism (Bhattacharya 2004), sex-biased differences in microhabitat selection, and differences in “catchability” or lags in emergence times of females relative to males (protandry) (Ehrlich et al. 1984; Frey and Leong 1993). In the case of adult *S. plana*, males and females differ significantly in their morphology and behavior. Males are less brightly-coloured and are much more competent flyers (Edwards 1994). By contrast, females are very cryptic and extremely poor dispersers (Edwards 1994). Females only display their bright orange hindwings to indicate their courtship solicitation during a very small fraction of their lifespans. Most of the time females are inactive (when males are patrolling) or are assiduous with the deposition of eggs; mainly during the afternoon when male activity has ceased. Also, adult male moths have been observed to be prey for robber flies (Asilidae family) and birds (Clarke and O’Dwyer 2000). Thus, adult males and females differ in their predation risks, catchability and visibility, which will greatly influence any evaluation of the adult sex ratio within populations. The sex ratio of pupal cases suggests that there is a natural bias in adult sex ratio, that somewhat mirrors that of the adults based on transect counts (Richter, personal observation). The occurrence in situ of empty lepidopteran pupal cases provided a specific and reliable indicator of habitat use by the species.

The presence of pupal cases in native grasslands and in grasslands that are comprised entirely of the exotic Chilean needle grass supports the view that this endemic Australian species has expanded its range to include these exotic

grasslands. However, the presence of pupal cases at sites that appear to be comprised entirely of Chilean needle grass does not necessarily indicate that the larvae have fed on the roots of this species. There is some possibility that the moths may have fed on the roots of residual species of native grass that are no longer evident in the grass sward, although this is becoming increasingly less likely as more sites are being found where the species appears to be breeding in Chilean needle grass (Gilmore et al. 2008; Richter et al. 2009; Brown et al. 2012). It is very likely that empty pupal cases can be used in molecular analyses to test for dietary content in *S. plana* [see for examples (Feinstein 2004; Watts et al. 2005)]. Stable isotope analysis is also now a widely used technique for investigating diet, trophic position and ecological function in communities (Post 2002) and, since it has been used with other lepidopteran pupal cases (Tibbets et al. 2008), it could be also used to investigate the feeding strategies in *S. plana*.

In conclusion, it is our view that non invasive sampling of pupal cases provides a valuable source of information about the critically endangered golden sun moth *S. plana*. The surveys can be undertaken readily by citizen scientists, providing an opportunity to contribute to resolving some of the remaining questions about the species conservation status and biology and establishing a much-needed monitoring program for the species. The pupal cases should be considered as being much more than “just an empty case” and hopefully will make an important contribution to future research and monitoring.

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